Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Python/Java distributed services**, **cloud infrastructure**, and **AI platforms**.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Restructured a fault-tolerant asynchronous **LLM orchestration** system with FastAPI background workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs while tripling throughput under concurrent workloads.
- Architected a multi-tenant platform on Azure SQL and Blob Storage with row-level isolation and indexed schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Designed a concurrent batch-processing pipeline using **multithreading** and secure PDF, DOCX, and ZIP ingestion through REST APIs and AWS RDS, processing 1K+ files per batch with fault-tolerant execution.
- Implemented a production AWS backend serving 150+ instructors and 10K+ daily grading requests, using CI/CD-driven rubric workflows to reduce feedback turnaround 60% while maintaining stable peak-load API responses.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer for offline reinforcement learning, improving long-horizon planning 18% in simulation while developing Python pipelines that processed 100K+ autonomous-driving trajectory logs through reproducible evaluation pipelines.
- Architected a spatial-temporal encoder joining HD-map geometry and ego-agent trajectories through multi-head attention, improving closed-loop planning stability 15% while comparing against three baselines in reproducible simulation experiments.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Troubleshot and optimized large-scale API integrations using object-oriented patterns, AWS Glue, and S3, improving response times 30%, pipeline efficiency 68%, and maintainability while reducing technical debt 20%.
- Engineered cross-platform microservices with Jenkins CI/CD and 20+ PyTest, Mockito, unit, and integration suites across Windows, Linux, and macOS while collaborating with Microsoft and GitHub stakeholders throughout Agile delivery.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Developed a MERN resource-management platform with eight **microservices**, REST APIs, and WebSockets, synchronizing 2M+ records across locations while reducing code-review turnaround 25% and sync conflicts 33% in production.
- Automated **Docker** builds, unit, integration, and end-to-end tests, and staged deployments through Jenkins CI/CD, achieving 95%+ validation while using Sentry, Azure Monitor, and Kibana for production observability.

Skills

Languages & Backend: Python, Java, C++, SQL, TypeScript, FastAPI, Flask, REST APIs, OOP, Data Structures
Infrastructure: Distributed Systems, Microservices, Concurrency, Multithreading, TCP/IP, Rate Limiting, Circuit Breakers, Feature Flags, Event Streaming
Data & Cloud: PostgreSQL, SQL Server, Azure SQL, SQLite, Redis, AWS, S3, Azure, GCP, Docker, Kubernetes
Reliability & AI: Observability, CI/CD, Git, GitHub Actions, Jenkins, PyTest, Performance Optimization, Monitoring, Sentry, Agile/Scrum, Code Reviews, LLM Systems, PyTorch

Projects

ControlMesh | *Python, FastAPI, Pydantic, SQLite, PyTest, GitHub Actions*

- Built a **Python/FastAPI** microservice control plane with versioned hot-reload configuration, token-bucket rate limits, circuit breakers, feature flags, durable event streams, and traffic replay; processed 10,000 public requests across 49 passing tests.

LoanLedger | *Java 21, Spring Boot, PostgreSQL, Event Sourcing, JUnit 5*

- Built an event-sourced ledger in **Java 21**, Spring Boot, and PostgreSQL with optimistic concurrency and exactly-once webhook handling, passing 5,500 accounting-invariant tests and 3,000 duplicate deliveries without double-posting.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Engineered a cloud-based data ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and performance tuning to reduce retrieval latency 40% for enterprise search workloads.